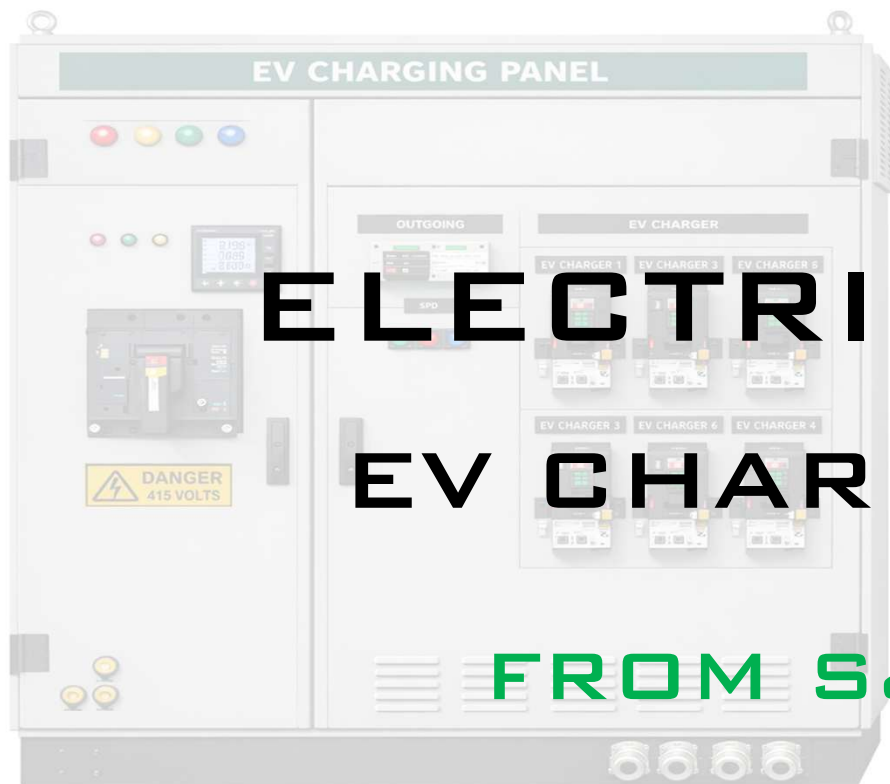




ELECTRICAL PANEL

EV CHARGING PANEL

FROM S&S GROUPS



Description About EV CHARGING PANEL

➤ **Design, Manufacturing, Supply, Installation, Testing & Commissioning (SITC) of EV Charging Distribution Panel** suitable for Electric Vehicle Charging infrastructure, fabricated out of high-quality CRCA sheet steel, suitable for indoor/outdoor installation as required, complete with all accessories, wiring, busbars, protection devices, earthing, and testing as per latest **IS / IEC standards**.

- Panel Construction
- Busbar System (Aluminium Type)
- Incoming Supply
- Outgoing Feeders
- Protection System
- Metering & Indication

- Internal Wiring
- Cable Entry
- Earthing
- Testing & Inspection
- Documentation
- Scope of Supply & Warranty



EV CHARGING PANEL

1. PANEL CONSTRUCTION

- ❑ The EV Charging Panel shall be fabricated from **2.0 mm thick CRCA sheet steel** for structure and **1.6 mm thick sheet for doors and covers**, properly reinforced with stiffeners.
- ❑ The panel shall be **floor mounted, front operated, cubicle type**, dust and vermin proof construction, finished with **7 tank anti-corrosive treatment and powder coated with epoxy polyester paint (minimum 60-80-micron thickness)**.

➤ THE PANEL SHALL INCLUDE :-

- ✓ Lockable front doors with neoprene gaskets
- ✓ Cable alley for incoming & outgoing cables
- ✓ Removable gland plates
- ✓ Proper ventilation louvers
- ✓ Danger notice plate
- ✓ Panel earthing studs
- ✓ Phase indication lamps (R-Y-B)
- ✓ Internal control wiring with FRLS copper wires

EV Charging Panel complete with MCCB / RCBO, MFM, SPD, Aluminum Busbars, Surge Protection, Cable Entry Glands with Plates etc.



2. BUSBAR SYSTEM (ALUMINUM TYPE)

- ❑ The Panel Shall Be Provided With **High Conductivity Ec Grade Aluminum Busbars** Of Adequate Rating.

➤ FEATURES INCLUDE :-

- ✓ Rated for suitable current capacity as per load requirement
- ✓ Fully insulated with **heat shrinkable PVC sleeves**
- ✓ Supported with **SMC / DMC insulated busbar supports**
- ✓ Designed to withstand **short circuit fault levels up to 50kA for 1 sec**
- ✓ Separate **Neutral Busbar** and **Earth Busbar** provided
- ✓ Busbars color coded for R-Y-B phases

**50KA FAULT
RATING**

**SMC / DMC
INSULATED
SUPPORTS**

NEUTRAL BUSBAR

3. INCOMING SUPPLY

- ❑ INCOMING POWER SUPPLY TO THE EV CHARGING PANEL SHALL BE CONTROLLED THROUGH :-
 - **MCCB / ACB (as per rating requirement)**
 - Thermal Magnetic protection
 - Adjustable overload and short circuit protection
 - Rotary operating handle with door interlock
- ❑ THE INCOMER SECTION SHALL INCLUDE :-
 - Digital multifunction meter (V, A, Hz, kW, kWh, PF)
 - CTs of required ratio
 - RYB phase indication lamps
 - Control MCBs

4. OUTGOING FEEDERS

- ❑ Outgoing feeders shall be provided for EV chargers using :-
 - **MCCB / MCB / RCBO protection** as required
 - Individual feeder protection for each EV charging point
 - Proper cable termination arrangement with lugs and glands
 - Ferruled control wiring

- ❑ Each feeder shall be clearly labeled with **engraved name plates**.

5. PROTECTION SYSTEM

❑ The EV Charging Panel shall include protection against :-

- Overload protection
- Short circuit protection
- Earth leakage protection
- Surge protection device (SPD) if required
- Under voltage / over voltage protection (optional)

6. METERING & INDICATION

❑ THE PANEL SHALL BE EQUIPPED WITH :-

- Digital multifunction energy meter
- Ammeter & Voltmeter with selector switch (if required)
- LED type phase indication lamps
- Control switches and indication for feeder status

7. INTERNAL WIRING

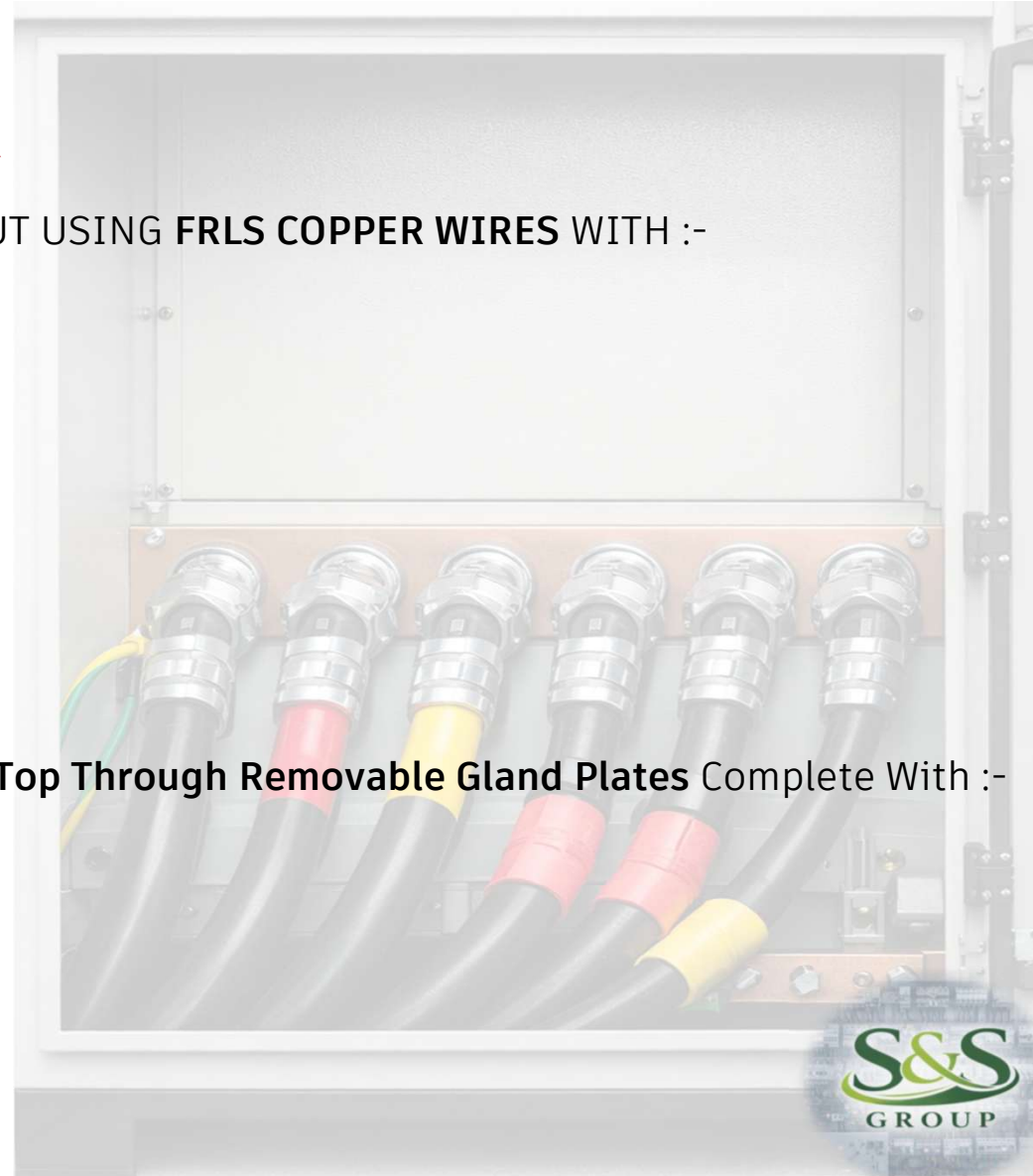
❑ ALL INTERNAL WIRING SHALL BE CARRIED OUT USING **FRLS COPPER WIRES** WITH :-

- PVC channels
- Ferrule numbering
- Neatly dressed wiring harness
- Terminal blocks for external connections

8. CABLE ENTRY

❑ Cable Entry Shall Be Provided From **Bottom / Top Through Removable Gland Plates** Complete With :-

- DOUBLE COMPRESSION CABLE GLANDS
- COPPER / ALUMINUM LUGS
- HEAT SHRINK SLEEVES



EARTHING

9. EARTHING

➤ THE PANEL SHALL BE PROVIDED WITH :-

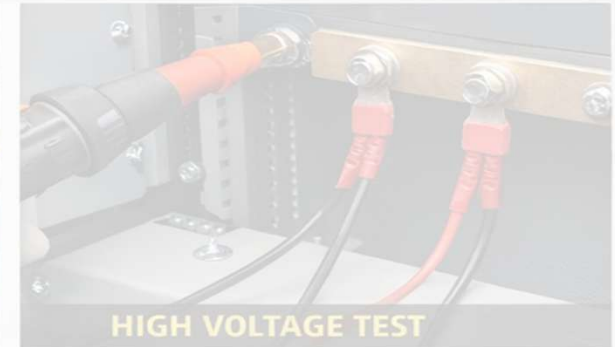
- ✓ **Two Separate Earthing Terminals**
- ✓ Proper Earthing Of Doors And Body Using Copper Flexible Links
- ✓ Earth Busbar Running Across The Panel

10. TESTING & INSPECTION

➤ THE PANEL SHALL BE TESTED AS PER **IS 8623 / IEC 61439 STANDARDS**, INCLUDING :

- ✓ Insulation Resistance Test
- ✓ High Voltage Test
- ✓ Functional Test Of Protection Devices
- ✓ Busbar Continuity Test

TESTING & INSPECTION



10. DOCUMENTATION

➤ The supplier shall provide :-

DOCUMENTATION

- ✓ GA drawing
- ✓ SLD diagram (Single Line Drawing)
- ✓ Wiring Diagram
- ✓ Test certificates
- ✓ Operation & maintenance manual

Test Certificates

Operation & Maintenance Manual

SCOPE INCLUDES

- ✓ Design & Engineering
- ✓ Fabrication & Assembly
- ✓ Supply of All Components
- ✓ Internal Wiring & Busbars
- ✓ Testing & Inspection
- ✓ Packing & Transportation

11. SCOPE INCLUDES & WARRANTY

➤ THE SCOPE SHALL INCLUDE :-

- ✓ Design & engineering
- ✓ Fabrication & Assembly
- ✓ Supply of all Components
- ✓ Internal wiring and Bus bar
- ✓ Testing & quality inspection
- ✓ Packing and transportation

WARRANTY :-

Minimum **12 months warranty** from date of commissioning.

